DATA MINING PRESENTATION

Clustering Concept & Algorithms

K-Means & K-Medoids Implementation

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What is Clustering?

Definition

Clustering is an **Unsupervised Learning** task that groups data points such that those in the same group are more similar to each other than to those in other groups.

It helps discover hidden patterns in unlabelled datasets without human intervention.

CLUSTERING VISUALIZATION

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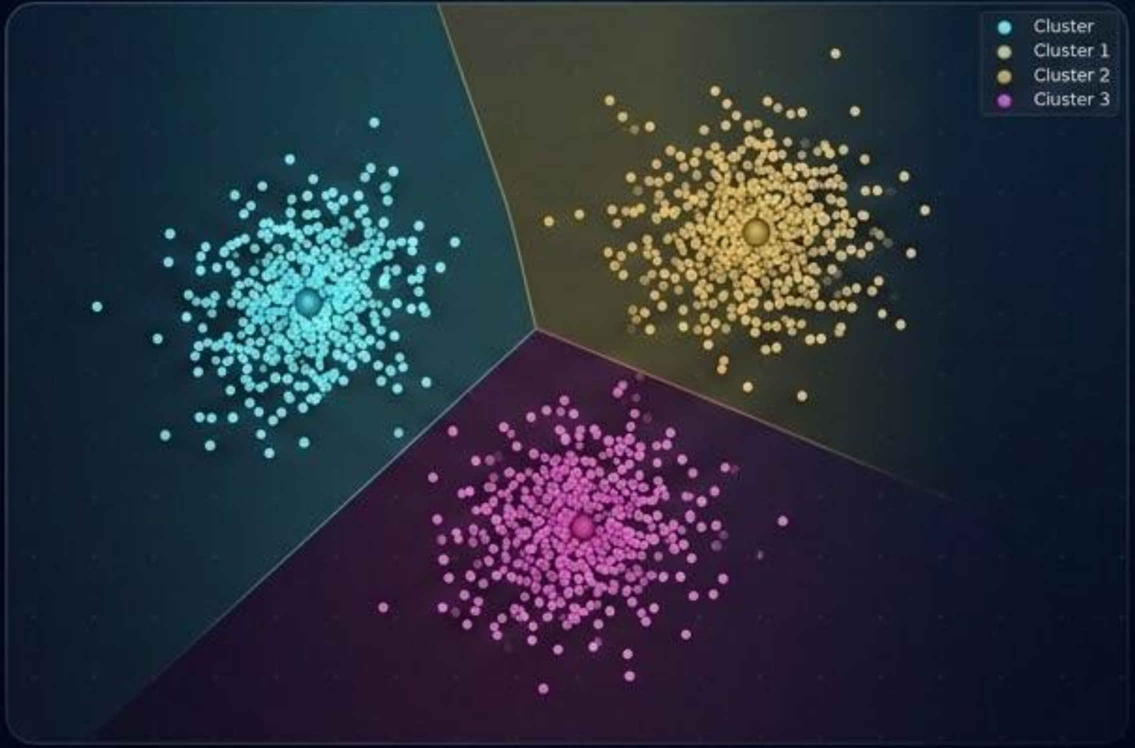


CLUSTERING VISUALIZATION

Core Objectives

Intra-cluster

Maximize similarity between points within the same cluster.

Inter-cluster

Minimize similarity between points in different clusters.

Discovery

Find natural structures and clusters in data.

Clustering Taxonomy

Partitioning

Constructs *k* partitions. (K-Means, K-Medoids)

Hierarchical

Creates a tree-like hierarchy of clusters.

Density-based

Based on connectivity and density functions.

Grid-based

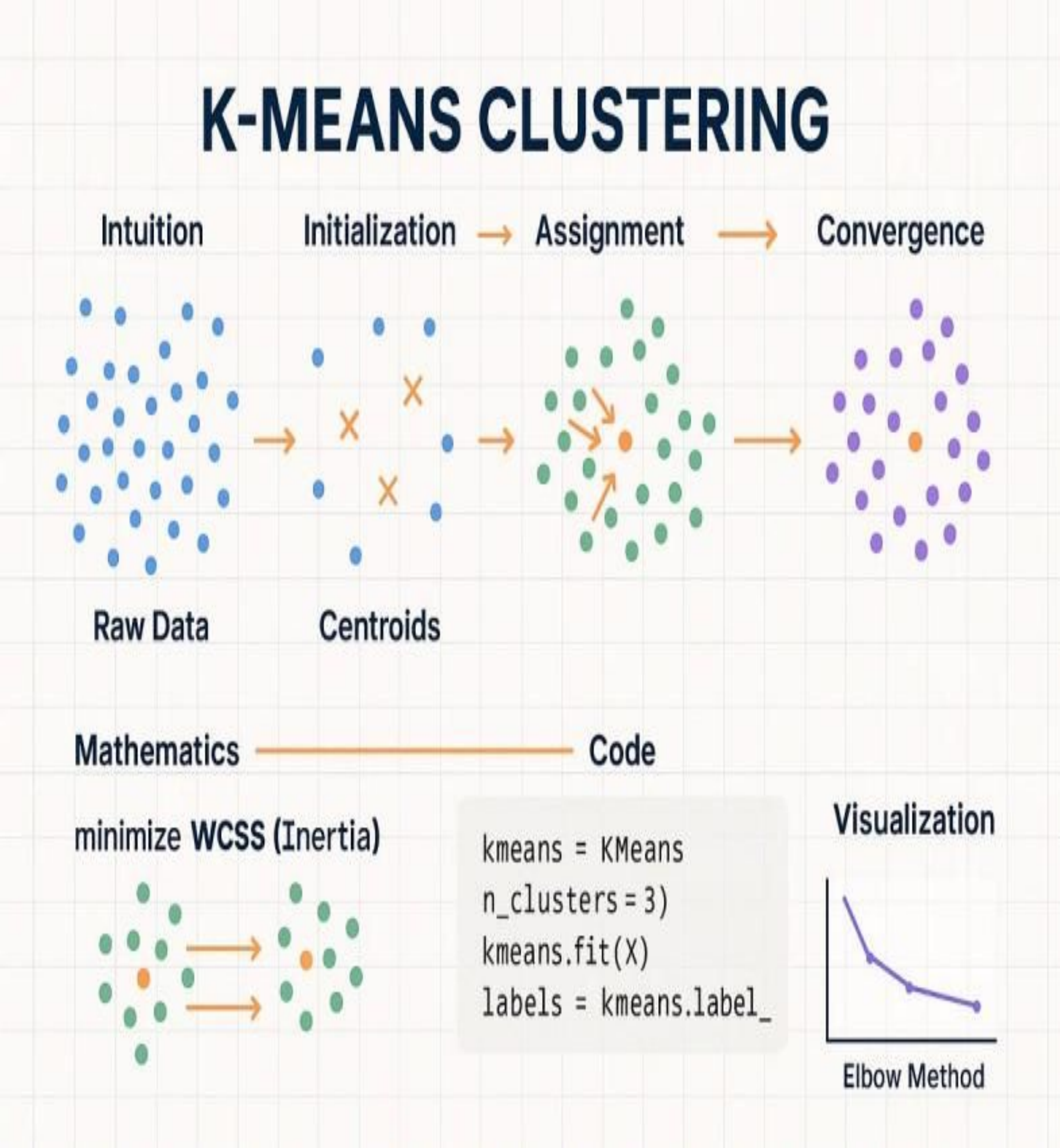
Based on a multiple-level granularity grid.

Centroid-based

K-Means defines clusters by their **Centroids** (mean values).

It is a centroid-based partitioning technique that minimizes the total within-cluster variance.

Goal: Reduce Squared Error.



The Algorithm Steps

1. Initialize: Pick *k* random centroids.

2. Assign: Allocate each point to the nearest centroid.

3. Update: Calculate the mean of each cluster as the new centroid.

4. Repeat: Iterate until centroids no longer move.

The Squared Error Function

K-Means seeks to minimize the Within-Cluster Sum of Squares (WCSS):

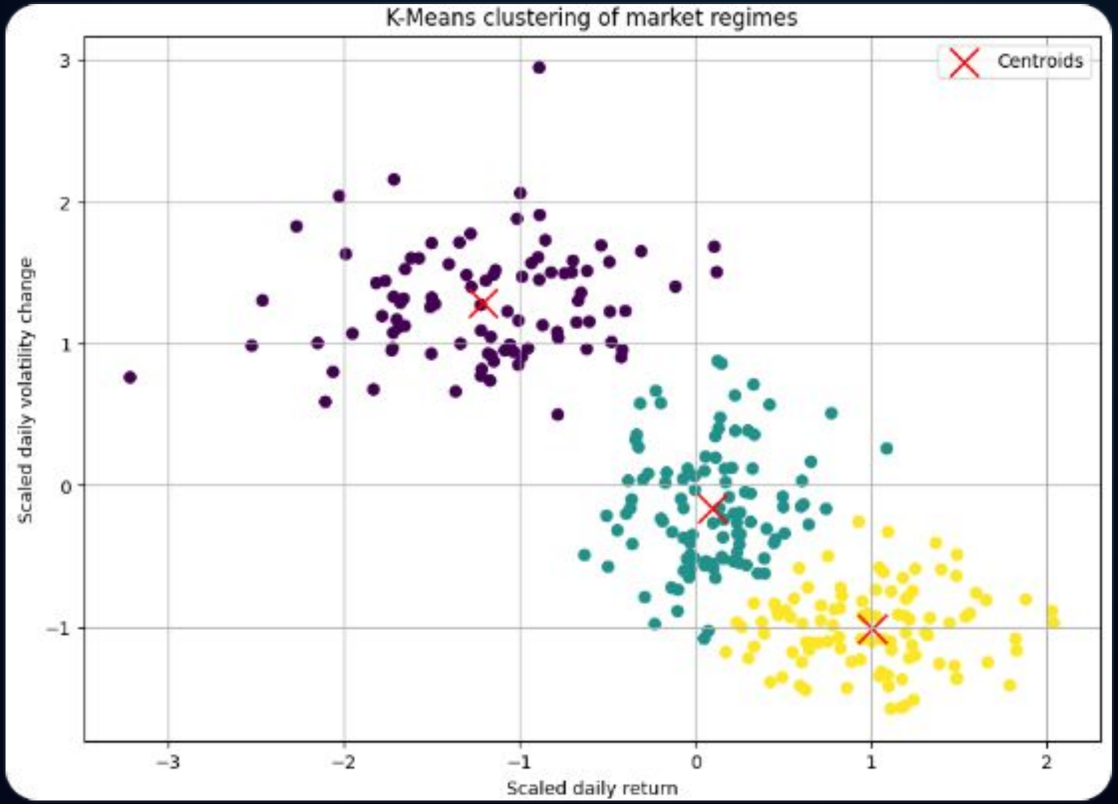
$$E = \sum_{i=1}^{k} \sum_{p \in C_i} ||p - m_i||^2$$

Where *mi* is the mean (centroid) of cluster *Ci*.

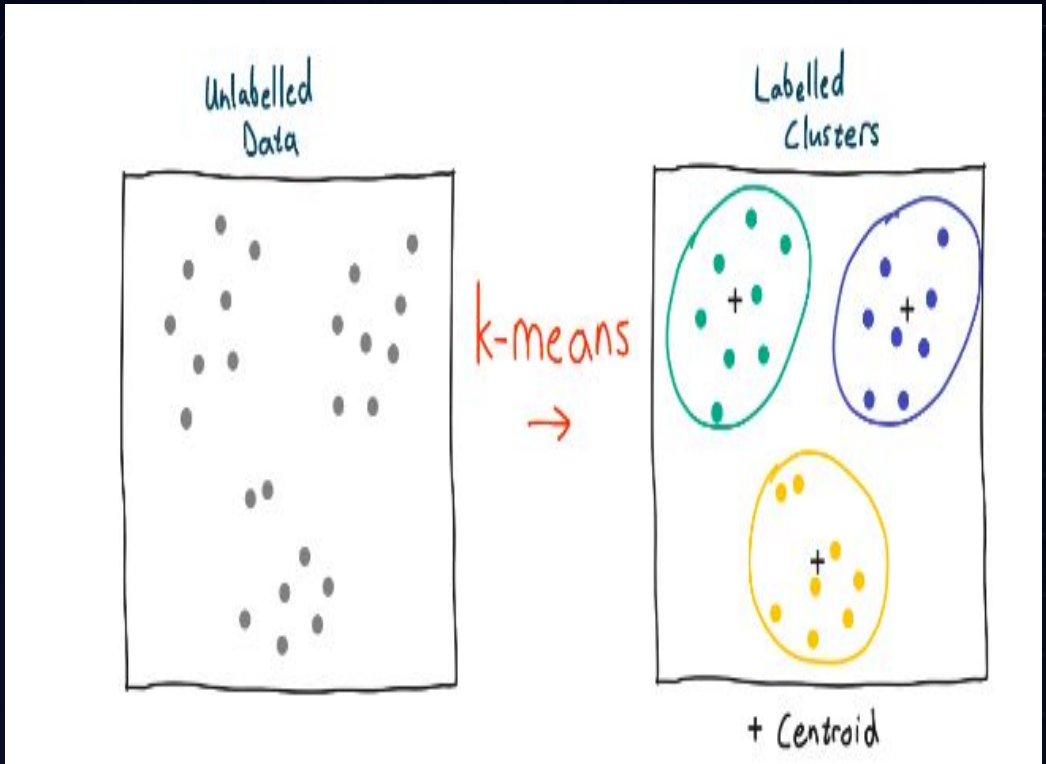
K-Means Limitations

Sensitivity to Outliers: Since it uses the mean, a single extreme point can pull the centroid away from the main cluster.

Noise Vulnerability: Works poorly with noisy data or non-spherical clusters.



Medoid Representation



Why use K-Medoids?

A **Medoid** is an actual data point that is the most centrally located object in a cluster.

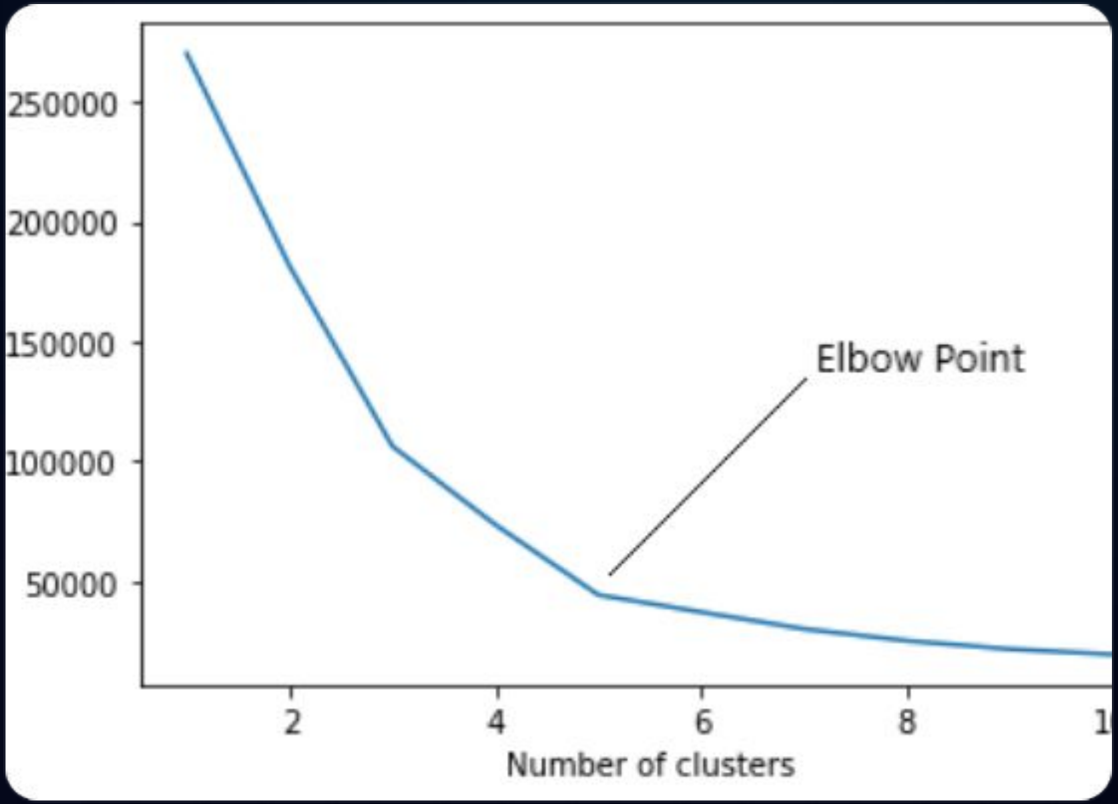
It is far more robust to outliers than K-Means because it uses an existing point rather than a calculated average.

K-Means vs K-Medoids

Feature	K-Means	K-Medoids (PAM)
Center type	Mean (Centroid)	Actual Point (Medoid)
Outlier sensitivity	High	Low (Robust)
Computational speed	Fast [O(nkt)]	Slower [O(k(n-k)2)]
Cluster Shape	Spherical	Arbitrary

The Elbow Method

We plot WCSS vs Number of clusters (*k*). The "Elbow" point indicates the optimal value where adding more clusters gives diminishing returns.



Thank you for your attention!

Data Mining Group | Bijendra, Subham, Sahan, Bongso

Image Sources

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